

DATA WRANGLING

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Outline

- Data wrangling
- Core packages in tidyverse
- Some common packages in tidyverse
- Descriptive statistics
- Measure of central tendency
- Skewness and measure of central tendency
- Mini exercise

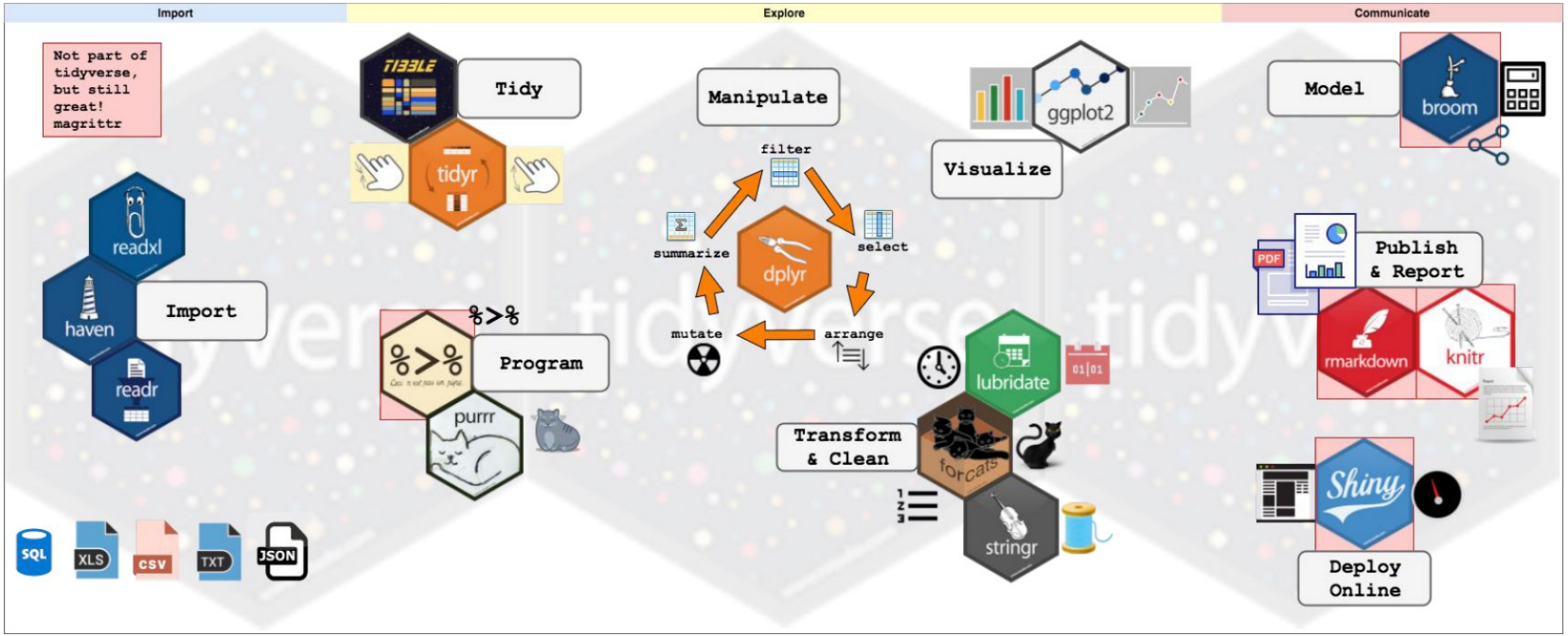
Data wrangling

- It is a process of getting the data into a useful form for visualisation, summary, and modeling/data analysis
- There are 2 approaches:
  - Tidyverse approach (we use this approach in this workshop)
  - Base R approach
- Tidyverse: a collection of R packages for data science, developed by the Posit team
  - 30 packages in total, 8 core packages; popular among R users, especially beginners

Core packages in tidyverse



Some common packages in tidyverse



Descriptive statistics

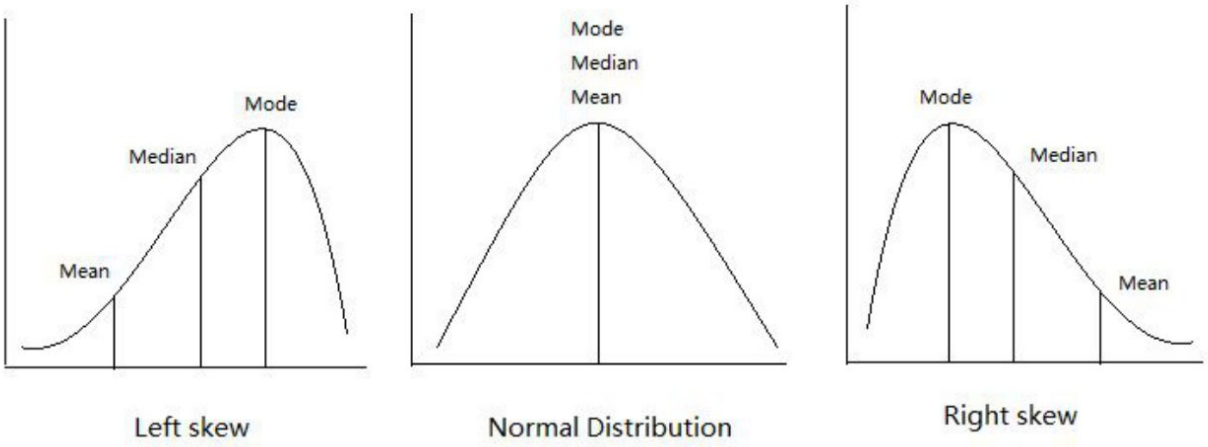
- Brief information that summarises the dataset
- Types
  - Measure of central tendency: mean, mode, median
  - Measure of variability: standard deviation, variance, interquartile range, minimum, maximum
- Usually mean (SD) for normal distribution and median (IQR) for non-normal distribution

Measure of central tendency

- Concerned with the average or middle value
- Mean: the average of the values
- Mode: the value that appears the most
- Median: the middle value in the ordered values

```
Data:   2, 3, 3, 4, 4, 4, 8
Mean:   (2+3+3+4+4+4+8) / 7 = 4
Median: middle value = 4
Mode:   appears the most = 4
```

Skewness and measures of central tendency



Mini exercise (15 min)

- Using the diabetes.csv dataset:
  a. Read the data and glimpse()
  b. filter() patients older than 50
  c. mutate() a new category variable (Age2 = Age * 2)
  d. group_by() + summarise() mean Age per group
  e. Drop the missing values and save the result in a new object



Any question?